Chat Export for sanidhya.rana_contractor@celanese.com

Assistant: Hello! I'm an AI assistant here to help you narrow down the Material requirements for Grade selection. Could you please tell me which application you're considering?

User: charging plug

Assistant: I understand this is for a charging plug application, likely related to an EV charging connector system.

1. What specific polymer component are you looking to make — the main plug housing, cable strain relief, grip overmold, internal insulator, seal, or another part?

2. Is this for standard AC charging or high-power DC fast charging? Also, will it be mainly used outdoors/public infrastructure or indoors/home charging?

3. Is the part single-material molded or does it involve insert molding/overmolding (for example soft grips, seals, or cable strain relief)?

User: it is an EV charging plug - cap and string

Assistant: Thanks, I understand the application is an EV charging plug protective cap with tether/string for the charging connector.

1. Is this used on a standard AC charging connector or a DC fast charging connector?

2. Is the cap and tether a single flexible molded part, or a multi-material assembly (for example rigid cap with soft tether/overmold)?

3. Do you have a target minimum wall thickness or any previous material currently being used for this part?

User: yes - it is strain relief. It is an outdoor charging station. we have used epdm before.

Assistant: Got it — this is an outdoor EV charging station application, and the cap/string area is functioning as a flexible strain-relief style component. I also captured that EPDM has been used previously.

1. Is the charging system standard AC charging or DC fast charging?

2. Is the part currently a single molded EPDM piece, or does it include overmolding/insert molding with a rigid section?

3. Do you have a minimum wall thickness or flexibility target for the strain-relief/tether area?

User: I am open to alternatives. no set thickness. it is a standalone molded part.

Assistant: Based on your EV charging plug cap and tether/strain-relief application, I captured these assumed requirements for confirmation:

1. Injection molded standalone flexible part

2. UV/weather resistance for outdoor charging station exposure

3. Hydrolysis/moisture resistance

4. High tear strength for tether durability

   - Assumed range: 92 to 175 kN/m²

5. Low water absorption

   - Assumed range: 0.005 to 11 %

6. Low humidity absorption

   - Assumed range: 0.002 to 1.3 %

7. Thermal shock resistance for outdoor temperature cycling

8. Heat stabilization for solar and elevated outdoor temperatures

9. Electrical insulation capability

   - Assumed electric strength: 21 to 39 kV/mm

10. Flexible/elastomeric behavior with strong flex-fatigue resistance for strain-relief function

11. Resistance to road salt, oils, rain, and cleaning chemicals

12. Long-term outdoor durability

Are all of these requirements correct for your application?

User: yes - I need f1. and UL 2551

Assistant: Understood. I captured F1 as the outdoor suitability/weatherability requirement along with the UL 2551 requirement for the cable/jacket-style flexible application environment.

A few remaining items to confirm:

1. Is flame retardancy required for this cap/tether component?

2. If flame retardancy is needed, do you have a target flame rating such as UL 94 V-0, V-1, or HB?

3. Does the cap need low compression set or sealing retention performance (for example maintaining a tight environmental seal over long-term outdoor use)?

User: i need a low shore hardness for assembly. I need F1 rating.

Assistant: You mentioned needing 'low shore hardness' for assembly. To ensure we match your requirement accurately, could you specify a numeric Shore A hardness value or range? For reference, Celanese materials offer Shore A hardness values from 25 (very soft) to 98 (very firm). Typically, 'low' would be in the 25–49 range. Please confirm if this range fits your needs, or provide a specific value if possible. If you're unsure, just let us know and we'll proceed with the general 'low' requirement.

User: yes- please use low... I need the lowest that can also meet f1

Assistant: I recommend connecting with a live Celanese materials specialist, because no flexible grade in the current result set fully confirms the complete combination of UL 2551 + UL F1 + very low hardness + outdoor electrical performance requirements.

Potential screening candidates for your EV charging plug cap/tether application are:

1. SANTOPRENE® 121-50M100

   - Best match for very low Shore hardness and elastomeric strain-relief behavior
   - Injection moldable
   - UV resistant
   - UL 746C F1 outdoor suitability support
   - Good fit for flexible assembly requirements
   - UL 2551 confirmation remains open

2. Hytrel® HTR8223 BK320

   - Good balance of flexibility and tear resistance
   - Strong flex-fatigue performance for tether applications
   - Low moisture absorption

- Heat/light stabilized
- Electrical strength below your target range and F1 not confirmed

3. Hytrel® HTR8341C BK320
   - Similar to HTR8223 BK320
   - Good tear strength and outdoor durability profile
   - Suitable for repeated flexing applications
   - F1 and UL 2551 not confirmed

4. Hytrel® HTR4275 BK600
   - Better documented outdoor durability profile
   - Good tear resistance
   - Injection moldable
   - Harder material than your preferred low-hardness target

5. Hytrel® HTR4275 BK316
   - Similar outdoor/weatherability profile to BK600
   - Better suited when durability is prioritized over softness
   - Hardness may be higher than desired for easy assembly

Key observations:

- SANTOPRENE® 121-50M100 appears to be the closest functional fit for the lowest hardness requirement.
- The Hytrel® grades provide stronger tear/flex durability but are generally firmer.
- UL 2551 confirmation is the main unresolved requirement across the current candidate set.

These grades may be suitable starting points for evaluation and validation. Final material selection, compliance verification, and product qualification remain your responsibility based on actual design and regulatory requirements.

Note: This recommendation is generated by an AI assistant and should be considered purely informational.